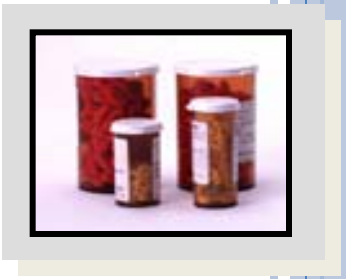
# Introduction to Probability and Statistics Fourteenth Edition

## Chapter 10 Inference from Small Samples

# INTRODUCTION



- When the sample size is small, the estimation and testing procedures of Chapter 8 are not appropriate.
- There are equivalent small sample test and estimation procedures for
  - ✓ $\mu$, the mean of a normal population
  - ✓ $\mu_1 - \mu_2$, the difference between two population means
  - ✓ $\sigma^2$, the variance of a normal population
  - ✓ The ratio of two population variances, $\sigma_1^2/\sigma_2^2$.

# General principle

Once we observe data: (1) we specify a model that can describe the data; (2) we focus on model parameters.

For a dataset, we do the following:

1. Model specification
2. Point estimtor(s)
3. Standardize the point estimator and identify its distribution
    1. Find the Confidence Interval
    2. Find the test statistics in a Hypothesis Test

# Models we have in this course

- No assumption on the model, but
  - $n$ is large: ch8, 9 Central Limit Theorem
    - $X \sim \mathcal{G}(\mu, \sigma^2)$ for some distribution $\mathcal{G}$
    - $X \sim Bernoulli(p)$
  - $n$ is small: ch15 Nonparametric statistics
- $n$ is small but $X \sim N(\mu, \sigma^2)$: ch 10

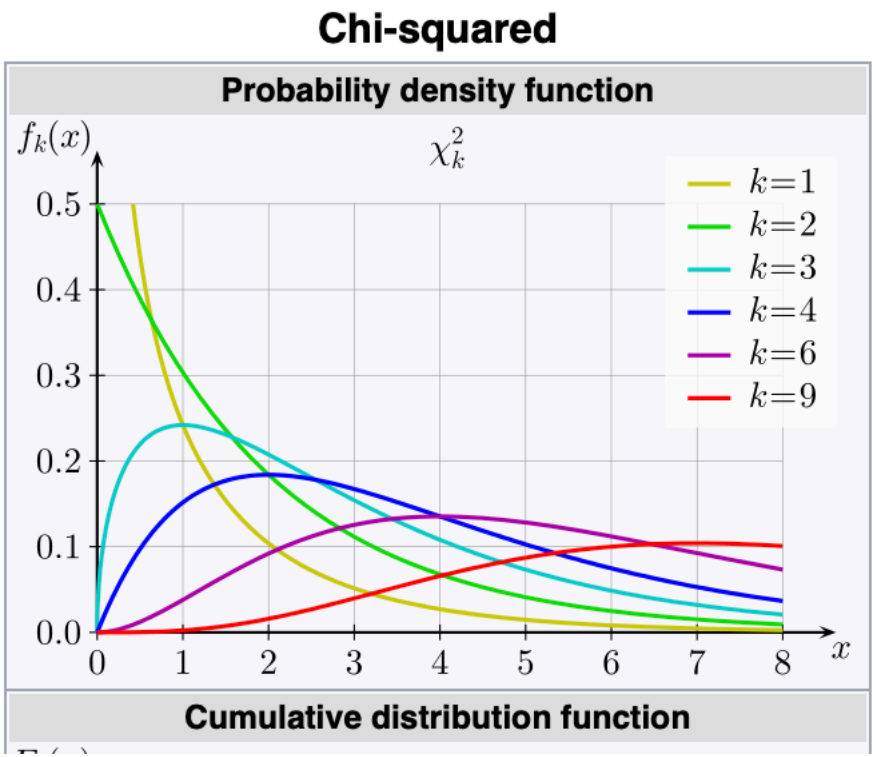
# 10.1 Student's *t* distribution

# Chi-squared distribution

If $Z_i \overset{i.i.d.}{\sim} N(0,1)$ for $i = 1,\ldots,m$, then

$$\sum_{i=1}^{m} Z_i^2 \sim \chi_m^2 .$$

$\chi_m^2$ is called the chi-squared distribution with degrees of freedom $m$.

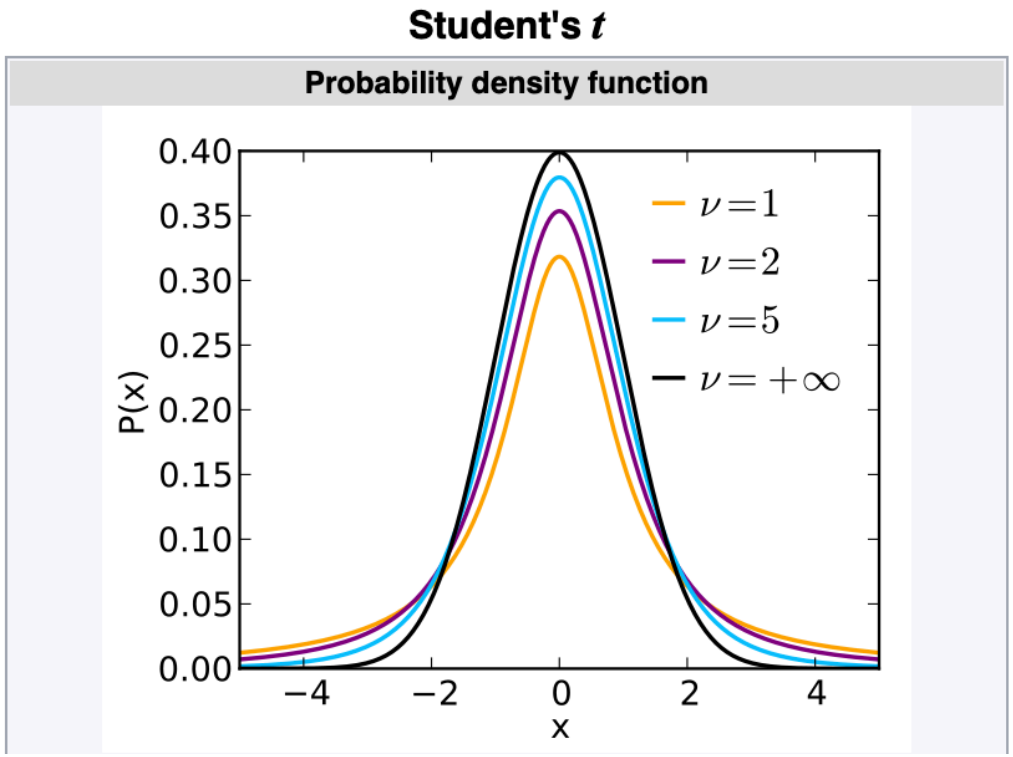


# Student's $t$ distribution

If $U \sim \chi^2_m$, $Z \sim N(0,1)$, and $U$ and $Z$ are independent, then

$$\frac{Z}{\sqrt{U/m}} \sim t_m .$$

$t_m$ is called the $t$ distribution with degrees of freedom $m$.



# Features of Student's $t$ distribution

1. Symmetric and mounded shaped.
2. When degrees of freedom goes to infinity, $t_m$ approaches the standard normal distribution.

Statistician William Sealy Gosset, known as "Student"

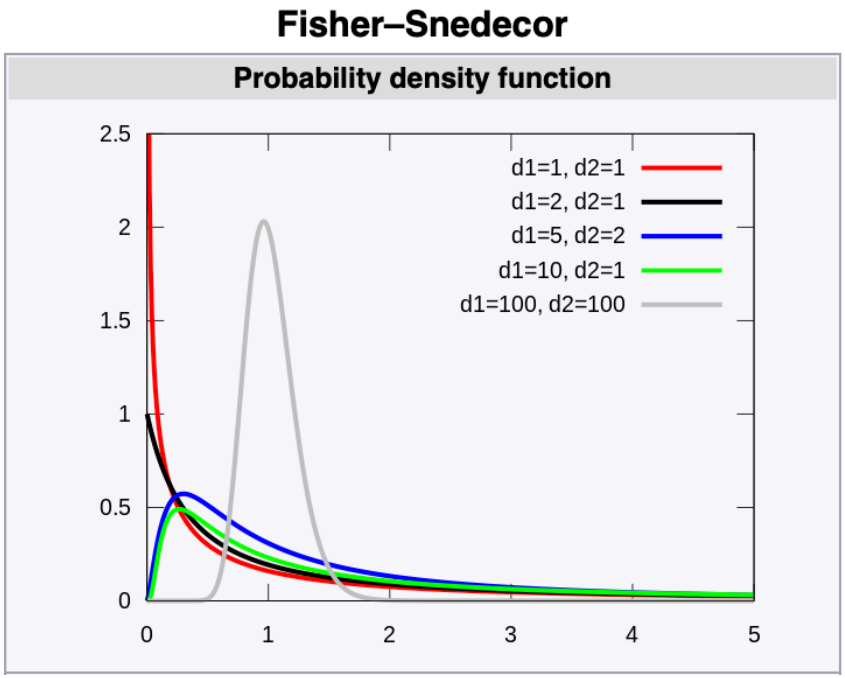
https://en.wikipedia.org/wiki/Student's_t-distribution

# $F$ DISTRIBUTION

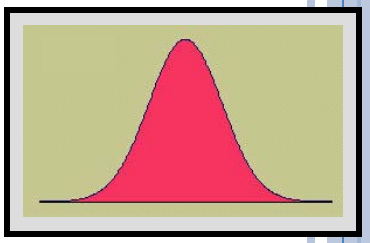
If $U \sim \chi^2_m$, $V \sim \chi^2_n$, and $U$ and $V$ are independent, then

$$\frac{U/m}{V/n} \sim F_{m,n}.$$

$F_{m,n}$ is called the $F$ distribution with degrees of freedom $m$ and $n$.



# The Sampling Distribution of the Sample Mean



- When we take a sample from a normal population, the sample mean $\bar{X}$ has a normal distribution for any sample size $n$, and

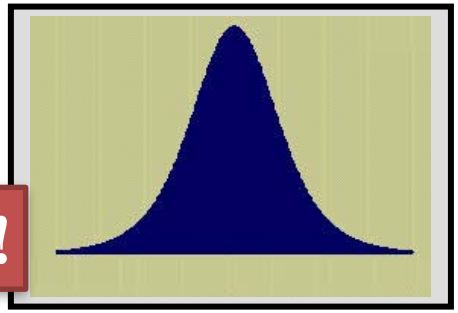
$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has a standard normal distribution.

- But if σ is unknown, and we must use $S$ to estimate it, the resulting statistic **is not normal**. $\frac{\bar{X} - \mu}{S/n}$ is not normal!

# Student's t Distribution

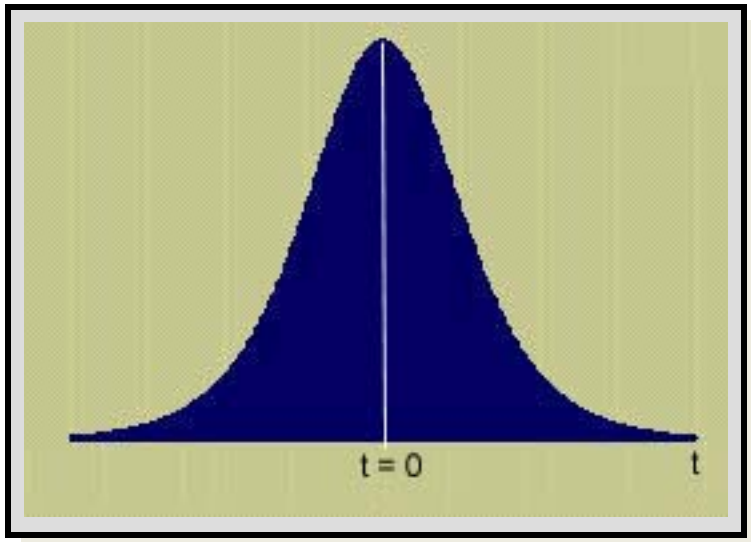
- Fortunately, this statistic does have a sampling distribution that is well known to statisticians, called the **Student's t distribution,** with ***n*-1 degrees of freedom.** $t = \dfrac{\bar{X} - \mu}{S/n} \sim t_{(n-1)}$



**Why? Go to Math Stat!**

- We can use this distribution to create estimation testing procedures for the population mean $\mu$.

# Properties of Student's *t*




- **Mound-shaped** and symmetric about 0.
- **More variable than *z***, with "heavier tails"

- Shape depends on the sample size $n$ or the **degrees of freedom, $n$-1.**
- As $n$ increases the shapes of the $t$ and $z$ distributions become almost identical.

MY APPLET

# STUDENT'S $t$ TABLI

- I use $t_{\nu;\alpha}$ to denote critical point of the distribution for the right-tailed probability of $\alpha$.
- $P(t_\nu > t_{\nu;\alpha}) = \alpha$ .

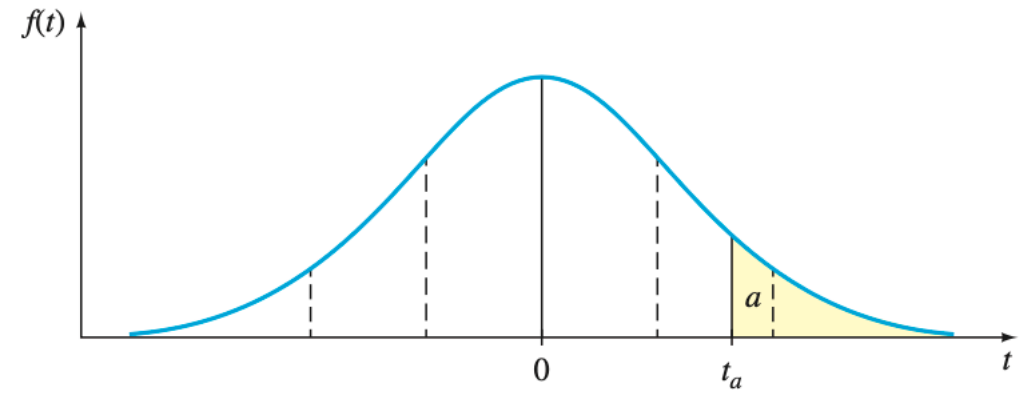



**Table 10.1 Format of the Student's $t$ Table from Table 4 in Appendix I**

| df | $t_{.100}$ | $t_{.050}$ | $t_{.025}$ | $t_{.010}$ | $t_{.005}$ | df |
|---|---|---|---|---|---|---|
| 1 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 | 1 |
| 2 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 | 2 |
| 3 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 | 3 |
| 4 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 | 4 |
| 5 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 | 5 |
| 6 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 | 6 |
| 7 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 | 7 |
| 8 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 | 8 |
| 9 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 | 9 |
| . | . | . | . | . | . | . |
| . | . | . | . | . | . | . |
| . | . | . | . | . | . | . |
| 28 | 1.313 | 1.701 | 2.048 | 2.467 | 2.763 | 28 |
| 29 | 1.311 | 1.699 | 2.045 | 2.462 | 2.756 | 29 |
| 30 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 | 30 |
| . | . | . | . | . | . | |
| . | . | . | . | . | . | . |
| . | . | . | . | . | . | . |
| 100 | 1.290 | 1.660 | 1.984 | 2.364 | 2.626 | 100 |
| 200 | 1.286 | 1.653 | 1.972 | 2.345 | 2.601 | 200 |
| 300 | 1.284 | 1.650 | 1.968 | 2.339 | 2.592 | 300 |
| 400 | 1.284 | 1.649 | 1.966 | 2.336 | 2.588 | 400 |
| 500 | 1.283 | 1.648 | 1.965 | 2.334 | 2.586 | 500 |
| inf. | 1.282 | 1.645 | 1.96 | 2.326 | 2.576 | inf. |

# USING THE *T*-TABLE

- Table 4 gives the values of $t$ that cut off certain critical values in the tail of the $t$ distribution.
- Index ***df*** and the appropriate tail area ***a*** to find $t_a$, the value of $t$ with area ***a*** to its right.

| *df* | $t_{.100}$ | $t_{.050}$ | $t_{.025}$ | $t_{.010}$ |
|---|---|---|---|---|
| 1 | 3.078 | 6.314 | 12.706 | 31.821 |
| 2 | 1.886 | 2.920 | 4.303 | 6.965 |
| 3 | 1.638 | 2.353 | 3.182 | 4.541 |
| 4 | 1.533 | 2.132 | 2.776 | 3.747 |
| 5 | 1.476 | 2.015 | 2.571 | 3.365 |
| 6 | 1.440 | 1.943 | 2.447 | 3.143 |
| 7 | 1.415 | 1.895 | 2.365 | 2.998 |
| 8 | 1.397 | 1.860 | 2.306 | 2.896 |
| 9 | 1.383 | 1.833 | 2.262 | 2.821 |
| 10 | 1.372 | 1.812 | 2.228 | 2.764 |
| 11 | 1.363 | 1.796 | 2.201 | 2.718 |
| 12 | 1.356 | 1.782 | 2.179 | 2.681 |
| 13 | 1.350 | 1.771 | 2.160 | 2.650 |
| 14 | 1.345 | 1.761 | 2.145 | 2.624 |
| 15 | 1.341 | 1.753 | 2.131 | 2.602 |

For a random sample of size $n = 10$, find a value of $t$ that cuts off .025 in the right tail.

Row = $df = n - 1 = 9$

Column subscript = $a = .025$

$t_{.025} = 2.262$

# Recall Ch 8 & 9

Suppose $X_i \overset{i.i.d.}{\sim} \mathcal{G}(\mu, \sigma^2)$ for $i = 1, \ldots, n$.

(1) When $\sigma$ is known, the Central Limit Theorem:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d.} N(0,1) \text{ as } n \to \infty$$

(2) When $\sigma$ is unknown, Slusky's theorem:

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \xrightarrow{d.} N(0,1) \text{ as } n \to \infty$$

$\xrightarrow{d.}$

- **Asymptotically distributed as 漸進分配**
- **Converges in distribution 分配收斂**

# Chapter 10

Suppose $X_i \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ for $i = 1, \ldots, n$.

(1) When $\sigma$ is known,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0,1)\,.$$

(2) When $\sigma$ is unknown,

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \sim t_{n-1}\,.$$

# 10.2 small-sample inferences concerning a population Mean

# EXAMPLE

A sprinkler system is designed so that the average time for the sprinklers to activate after being turned on is no more than 15 seconds. A test of 6 systems gave the following times:

17, 31, 12, 17, 13, 25

Is the system working as specified? Test using $\alpha = .05$.

$H_0 : \mu = 15$ (working as specified)

$H_a : \mu > 15$ (not working as specified)

# Key ingredients (when $\sigma$ is unknown)

1. Model: $X_i \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ and we focus on the estimation of $\mu$.
2. Point estimator of $\mu$: $\bar{X}$
3. If $\sigma^2$ is unknown, we use the sample variance $S^2$ to estimate it. Then, the *standardized point* estimator and its distribution: $\dfrac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}$.

# Confidence Interval

- $$1-\alpha = P(-t_{(n-1);\alpha/2} < \frac{\bar{X}-\mu}{S/\sqrt{n}} < t_{(n-1);\alpha/2})$$
$$= P(-t_{(n-1);\alpha/2}\frac{S}{\sqrt{n}} < \bar{X}-\mu < t_{(n-1);\alpha/2}\frac{S}{\sqrt{n}})$$
$$= P(-t_{(n-1);\alpha/2}\frac{S}{\sqrt{n}} < \mu-\bar{X} < t_{(n-1);\alpha/2}\frac{S}{\sqrt{n}})$$
$$= P(\bar{X}-t_{(n-1);\alpha/2}\frac{S}{\sqrt{n}} < \mu < \bar{X}+t_{(n-1);\alpha/2}\frac{S}{\sqrt{n}})$$
- With data, the $100(1-\alpha)\%$ CI for the population mean $\mu$ as $\bar{X} \pm t_{(n-1);\alpha/2}\frac{S}{\sqrt{n}}$.

# Hypothesis test

1. $H_0 : \mu = \mu_0$ versus $H_a : \mu \neq \mu_0$ (or > or <)
2. Set the value of $\alpha$.
3. The test statistic and its sampling distribution is

$$t_{STAT} = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{(n-1)}.$$

4. Calculate the realized statistic, $t^* = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$.
5. Find the rejection region or $p$-value.
6. Conclude.
   - Reject or do not reject $H_0$.
   - Conclude in plain language and get back to the scenario.

# EXAMPLE

**Data:** 17, 31, 12, 17, 13, 25

First, calculate the sample mean and standard deviation, using your calculator or the formulas in Chapter 2.

$$\bar{x} = \frac{\sum x_i}{n} = \frac{115}{6} = 19.167$$

$$s = \sqrt{\frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}} = \sqrt{\frac{2477 - \frac{115^2}{6}}{5}} = 7.387$$

# EXAMPLE

**Data:** 17, 31, 12, 17, 13, 25

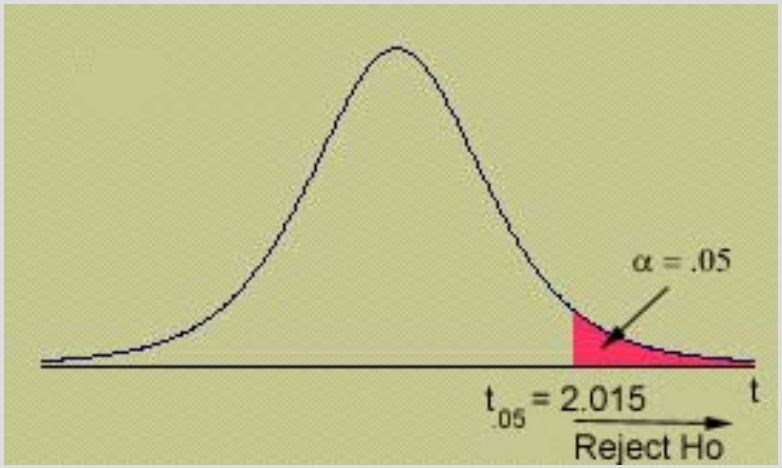
Calculate the test statistic and find the rejection region for α =.05.

Test statistic :

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{19.167 - 15}{7.387/\sqrt{6}} = 1.38$$

Degrees of freedom :

$$df = n - 1 = 6 - 1 = 5$$




**Rejection Region:** Reject $H_0$ if $t > 2.015$. If the test statistic falls in the rejection region, its *p*-value will be less than α = .05.

# Conclusion

**Data:** 17, 31, 12, 17, 13, 25
Compare the observed test statistic to the rejection region, and draw conclusions.

$$H_0 : \mu = 15$$
$$H_a : \mu > 15$$

Test statistic : $t = 1.38$

Rejection Region :

Reject $H_0$ if $t > 2.015$.

**Conclusion:** For our example, $t$ = 1.38 does not fall in the rejection region and $H_0$ is not rejected. There is insufficient evidence to indicate that the average activation time is greater than 15.

# Statistical conclusion (careful!)

- Because realized statistic does not fall into the rejection region, so we do not reject $H_0$.
- Do not reject $H_0$
- Insignificance evidence to show $H_0$ is incorrect $\approx$ Insignificant evidence to show that $H_a$ is correct

# SOP with Rejection Region Approach

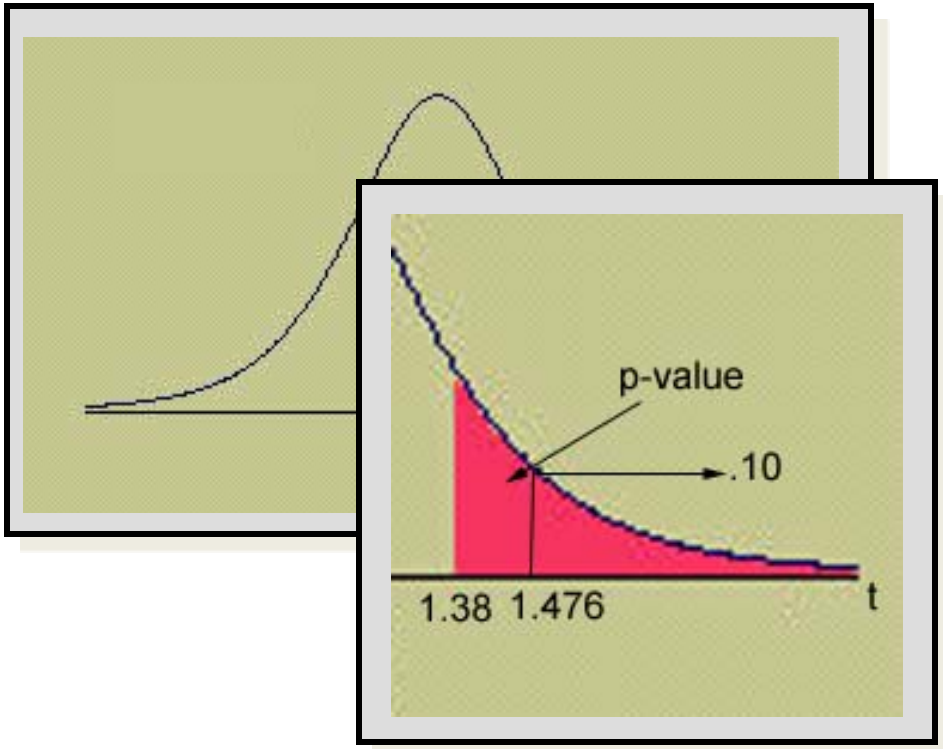
1. $H_0 : \mu = 15$ versus $H_a : \mu > 15$
2. Set $\alpha = 0.05$.
3. Define the test statistic $t_{STAST} = \frac{\bar{X} - 15}{S/\sqrt{n}} \sim t_{(6-1)}$.
4. Calculate the realized test statistic

$t^* = 1.38$.

5. Rejection region $\{t : t > 2.015\}$.
6. Because $t^*$ does not fall into the rejection region, we do not reject $H_0$.

# APPROXIMATING THE *P*-VALUE

- You can only approximate the *p*-value for the test using Table 4.




| df | $t_{.100}$ | $t_{.050}$ |
|---|---|---|
| 1 | 3.078 | 6.314 |
| 2 | 1.886 | 2.920 |
| 3 | 1.638 | 2.353 |
| 4 | 1.533 | 2.132 |
| 5 | 1.476 | 2.015 |

Since the observed value of $t$ = 1.38 is smaller than $t_{.10}$ = 1.476,

*p*-value > .10.

# $p$-VALUE APPROACH

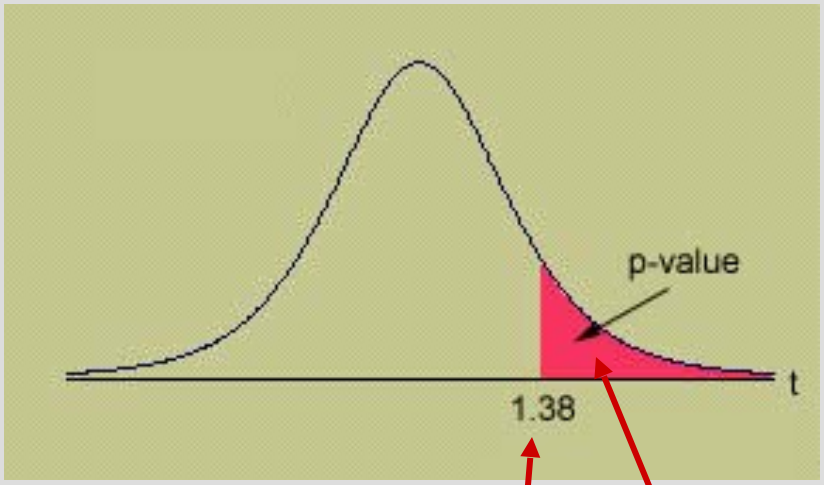
5. The p-value > 0.10 > $\alpha = 0.05$
6. We do not rejection $H_0$.

MY APPLET

# THE EXACT *P*-VALUE

- You can get the exact *p*-value using some calculators or a computer.

*p*-value = .113 which is greater than .10 as we approximated using Table 4.




**One-Sample T: Times**

Test of mu = 15 vs > 15

| Variable | N | Mean | StDev | SE Mean | 95% Lower Bound | T | P |
|---|---|---|---|---|---|---|---|
| Times | 6 | 19.1667 | 7.3869 | 3.0157 | 13.0899 | 1.38 | 0.113 |

# 10.3

## Small sample inference for the difference between two population means: independent random sample

# EXAMPLE

- Two training procedures are compared by measuring the time that it takes trainees to assemble a device. A different group of trainees are taught using each method. Is there a difference in the two methods? Use $\alpha$ = .01.

| Time to Assemble | Method 1 | Method 2 |
|---|---|---|
| Sample size | 10 | 12 |
| Sample mean | 35 | 31 |
| Sample Std Dev | 4.9 | 4.5 |

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_a : \mu_1 - \mu_2 \neq 0$$

Test statistic :

$$t = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

# Recall

- A linear combination of normal distribution remains a normal distribution:
- $$(\bar{X}_1 - \bar{X}_2) \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

# Key ingredients

1. Model

$$X_{1i} \sim N(\mu_1, \sigma_1^2), \text{ for } i = 1, \ldots, n_1$$
$$X_{2j} \sim N(\mu_2, \sigma_2^2), \text{ for } j = 1, \ldots, n_2$$

2. Point estimator of $(\mu_1 - \mu_2)$: $(\bar{X}_1 - \bar{X}_2)$.
3. If $\sigma_1^2$ and $\sigma_2^2$ are known, the normalized point estimator and its sampling distribution:

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim Z.$$

# $\sigma_1^2$ AND $\sigma_2^2$ ARE KNOWN

We estimate them using sample variances $S_1^2$ and $S_2^2$:

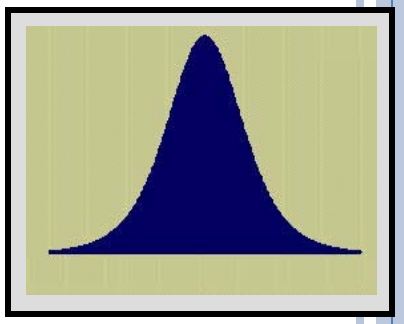
$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)^2$$

$$S_2^2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (X_{2j} - \bar{X}_2)^2$$

Then, we furthermore consider two cases:

1. $\sigma_1^2 = \sigma_2^2$
2. $\sigma_1^2 \neq \sigma_2^2$

# Testing the Difference between Two Means



- How can you tell if the equal variance assumption is reasonable?

**Back in chapter 10.6!**

Rule of Thumb:

If the ratio, $\dfrac{\text{larger } s^2}{\text{smaller } s^2} \le 3$,

the equal variance assumption is reasonable.

If the ratio, $\dfrac{\text{larger } s^2}{\text{smaller } s^2} > 3$,

use an alternative test statistic.

# Case I: $\sigma_1^2 = \sigma_2^2$. *POOLED METHOD*

Assume $\sigma_1^2$ and $\sigma_2^2$ are unknown but they are equal

Define the common sample variance by

$$S^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

Then, we have

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S^2(\frac{1}{n_1} + \frac{1}{n_2})}} \sim t_{(n_1+n_2-2)}$$

# CI

Follow the same manner:

$$1-\alpha = P\left(-t_{(n_1+n_2-2);\alpha/2} < \frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sqrt{S^2(\frac{1}{n_1}+\frac{1}{n_2})}} < t_{(n_1+n_2-2);\alpha/2}\right)$$

$$= P\left(-t_{(n_1+n_2-2);\alpha/2}\sqrt{S^2(\frac{1}{n_1}+\frac{1}{n_2})} < (\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2) < t_{(n_1+n_2-2);\alpha/2}\sqrt{S^2(\frac{1}{n_1}+\frac{1}{n_2})}\right)$$

$$= P\left((\bar{X}_1-\bar{X}_2)-t_{(n_1+n_2-2);\alpha/2}\sqrt{S^2(\frac{1}{n_1}+\frac{1}{n_2})} < (\mu_1-\mu_2) < (\bar{X}_1-\bar{X}_2)+t_{(n_1+n_2-2);\alpha/2}\sqrt{S^2(\frac{1}{n_1}+\frac{1}{n_2})}\right).$$

The $100(1-\alpha)\,\%$ confidence interval for $(\mu_1-\mu_2)$ is

$$(\bar{X}_1-\bar{X}_2) \pm t_{(n_1+n_2-2);\alpha/2}\sqrt{S^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}.$$

# Hypothesis test

3. For $H_0 : \mu_1 - \mu_2 = D_0$, the test statistic and its sampling distribution is

$$t_{STAT} = \frac{(\bar{X}_1 - \bar{X}_2) - D_0}{\sqrt{S^2(\frac{1}{n_1} + \frac{1}{n_2})}} \sim t_{(n_1+n_2-2)}\,.$$

4. The realized statistic is

$$t^* = \frac{(\bar{x}_1 - \bar{x}_2) - D_0}{\sqrt{s^2(\frac{1}{n_1} + \frac{1}{n_2})}}\,.$$

MY APPLET

# EXAMPLE

- Solve this problem by approximating the *p*-value using Table 4.

| Time to Assemble | Method 1 | Method 2 |
|---|---|---|
| Sample size | 10 | 12 |
| Sample mean | 35 | 31 |
| Sample Std Dev | 4.9 | 4.5 |

Calculate:

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{9(4.9^2) + 11(4.5^2)}{20} = 21.942$$

Test statistic:

$$t = \frac{35 - 31}{\sqrt{21.942\left(\frac{1}{10} + \frac{1}{12}\right)}}$$

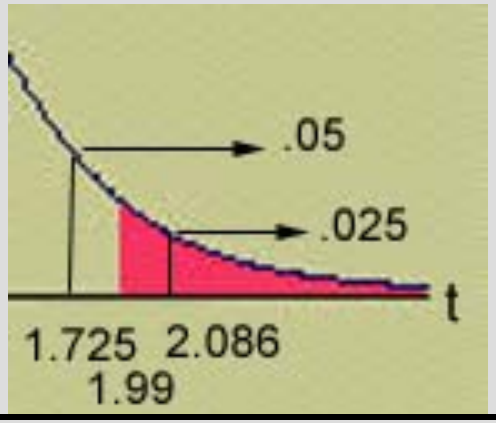
$$= 1.99$$

# EXAMPLE

$$p\text{-value}: P(t > 1.99) + P(t < -1.99)$$

$$P(t > 1.99) = \frac{1}{2}(p\text{-value})$$

$df = n_1 + n_2 - 2 = 10 + 12 - 2 = 20$




| df | $t_{.100}$ | $t_{.050}$ | $t_{.025}$ | $t_{.010}$ | $t_{.005}$ | df |
|---|---|---|---|---|---|---|
| 19 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 | 19 |
| 20 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 | 20 |

.025 < ½( *p*-value) < .05

.05 < *p*-value < .10

Since the *p*-value is greater than $\alpha$ = .01, $H_0$ is not rejected. There is insufficient evidence to indicate a difference in the population means.

# SOP WITH P-VALUE APPROACH

1. $H_0 : \mu_1 - \mu_2 = 0$ versus $H_a : \mu_1 - \mu_2 \neq 0$.
2. $\alpha = 0.01$.
3. Test statistic and its sampling distribution

$$t_{STAT} = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{\sqrt{S^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{(n_1+n_2-2)=20}\,.$$

4. Realized test statistic $t^* = 1.99$.
5. Approximate p-value:
   $0.05 = 2 \times 0.025 < p\text{−value} < 2 \times 0.05 = 0.10.$
6. Because $p - value > 0.01$, we do not reject $H_0$.

## Case 2: $\sigma_1^2 \neq \sigma_2^2$. *Satterthwaite's approximation*

Assume $\sigma_1^2$ and $\sigma_2^2$ are unknown, but $\sigma_1^2 \neq \sigma_2^2$.

3. The standardized point estimator for $(\mu_1 - \mu_2)$ and its sampling distribution is

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim t_\nu,$$

with

$$\nu \approx \text{floor}\left( \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \right).$$

We will *round down* to the nearest integer.

## Case 2: $\sigma_1^2 \neq \sigma_2^2$. *Satterthwaite's approximation*

The CI is $(\bar{x}_1 - \bar{x}_2) \pm t_{\nu;\alpha/2}\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$.

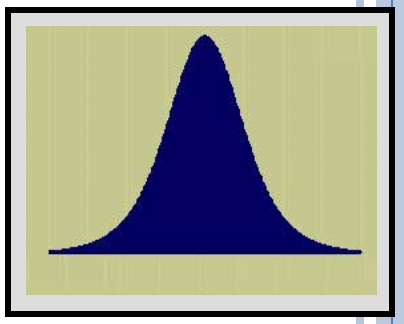
For $H_0 : \mu_1 - \mu_2 = D_0$, the test statistic and its sampling distribution is

$$\frac{(\bar{X}_1 - \bar{X}_2) - D_0}{\sqrt{\dfrac{S_1^2}{n_1} + \dfrac{S_2^2}{n_2}}} \sim t_\nu .$$

# 10.4

## Small-sample inferences for the difference between two population means: A paired difference test

# The Paired-Difference Test



- Sometimes the assumption of independent samples is intentionally violated, resulting in a **matched-pairs** or **paired-difference test**.
- By designing the experiment in this way, we can eliminate unwanted variability in the experiment by analyzing only the differences,

$$d_i = x_{1i} - x_{2i}$$

- to see if there is a difference in the two population means, $\mu_1 - \mu_2$.

# Example

| Car | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Type A | 10.6 | 9.8 | 12.3 | 9.7 | 8.8 |
| Type B | 10.2 | 9.4 | 11.8 | 9.1 | 8.3 |

- One Type A and one Type B tire are randomly assigned to each of the rear wheels of five cars. Compare the average tire wear for types A and B using a test of hypothesis.

$$H_0 : \mu_1 - \mu_2 = 0$$
$$H_a : \mu_1 - \mu_2 \neq 0$$

- But the samples are not independent. The pairs of responses are linked because measurements are taken on the same car.

# Example

| Car | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Type A | 10.6 | 9.8 | 12.3 | 9.7 | 8.8 |
| Type B | 10.2 | 9.4 | 11.8 | 9.1 | 8.3 |

- One Type A and one Type B tire are randomly assigned to each of the rear wheels of five cars. Compare the average tire wear for types A and B using a test of hypothesis.

- But the samples are not independent. The pairs of responses are linked because measurements are taken on the same car.

# Example

| Car | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Type A | 10.6 | 9.8 | 12.3 | 9.7 | 8.8 |
| Type B | 10.2 | 9.4 | 11.8 | 9.1 | 8.3 |
| Difference | .4 | .4 | .5 | .6 | .5 |

# Key ingredients

1. Model: $X_{1i} - X_{2i} \sim N(\mu_1 - \mu_2, \sigma^2)$, for $i = 1, \ldots, n$.

   Define $D_i = X_{1i} - X_{2i}$. The model is

$$D_i \sim N(\mu_1 - \mu_2, \sigma^2), \text{ for } i = 1, \ldots, n.$$

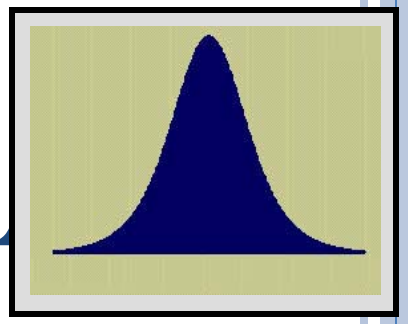
2. Point estimator for $(\mu_1 - \mu_2)$: $\bar{D} = \frac{1}{n}\sum_{i=1}^{n} D_i$

3. Standardized estimator and its distribution:

$$\frac{\bar{D} - (\mu_1 - \mu_2)}{S_D/\sqrt{n}} \sim t_{(n-1)},$$

   where $S_D = \sqrt{\frac{\sum_{i=1}^{n}(D_i - \bar{D})^2}{n-1}}$

# Confidence Interval



Follow the same manner

$$1-\alpha = P\left(-t_{(n-1);\alpha/2} < \frac{\bar{D}-(\mu_1-\mu_2)}{S_D/\sqrt{n}} < t_{(n-1);\alpha/2}\right)$$

$$= P\left(-t_{(n-1);\alpha/2}\frac{S_D}{\sqrt{n}} < \bar{D}-(\mu_1-\mu_2) < t_{(n-1);\alpha/2}\frac{S_D}{\sqrt{n}}\right)$$

$$= P\left(\bar{D}-t_{(n-1);\alpha/2}\frac{S_D}{\sqrt{n}} < (\mu_1-\mu_2) < \bar{D}+t_{(n-1);\alpha/2}\frac{S_D}{\sqrt{n}}\right)$$

You can construct a 100(1-$\alpha$)% confidence interval for a paired experiment using

$$\bar{d} \pm t_{(n-1),\alpha/2}\frac{s_d}{\sqrt{n}}.$$

•

# Hypothesis test

1. $H_0 : \mu_1 - \mu_2 = D_0$
2. Set $\alpha$
3. the test statistic and its sampling distribution is

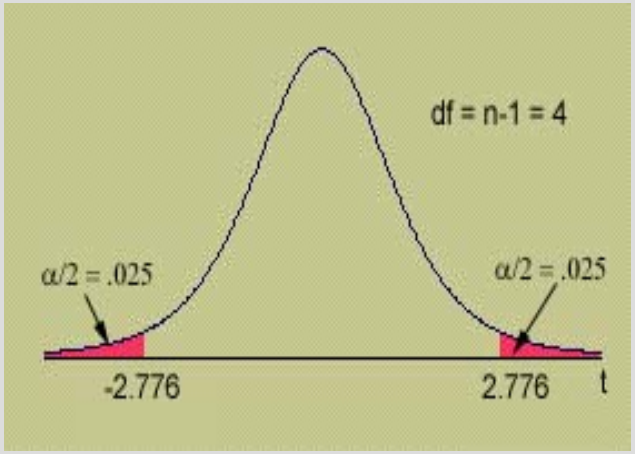
$$t_{STAT} = \frac{\bar{D} - D_0}{S_D/\sqrt{n}} \sim t_{(n-1)}$$

4. The realized test statistic $t^* = \dfrac{\bar{d} - D_0}{s_d/\sqrt{n}}$
5. Rejection rejection or p-value approach
6. Conclude.

# EXAMPLE

Find the 95 % CI for the average difference of tire wear between types A and B!

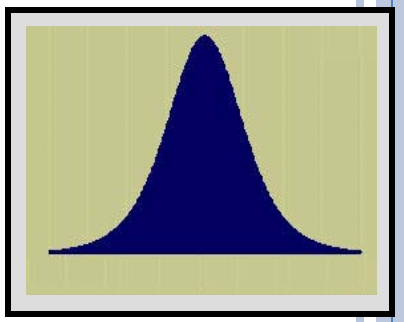
| Car | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Type A | 10.6 | 9.8 | 12.3 | 9.7 | 8.8 |
| Type B | 10.2 | 9.4 | 11.8 | 9.1 | 8.3 |
| Difference | .4 | .4 | .5 | .6 | .5 |




$$\text{Calculated } \bar{d} = \frac{\sum d_i}{n} = .48$$

$$s_d = \sqrt{\frac{\sum d_i^2 - \frac{\left(\sum d_i\right)^2}{n}}{n-1}} = .0837$$

# CI



- Because $\bar{d} = 0.48$, $s_d = 0.0837$, and $t_{4,0.025} = 2.776$, the 95% CI is

$$\bar{d} \pm t_{(n-1);\alpha/2}\frac{s_d}{\sqrt{n}} = \left[0.48 \pm 2.776\frac{0.0837}{\sqrt{5}}\right] = [0.37, 0.58]$$

- In repeated sampling, about 95 % of intervals constructed this way will cover the true average difference of the tire wear between types A and B.
- There seems a difference of tire wear between types A and B.

# EXAMPLE

| Car | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Type A | 10.6 | 9.8 | 12.3 | 9.7 | 8.8 |
| Type B | 10.2 | 9.4 | 11.8 | 9.1 | 8.3 |

- One Type A and one Type B tire are randomly assigned to each of the rear wheels of five cars. Compare the average tire wear for types A and B using a test of hypothesis.

$$H_0 : \mu_1 - \mu_2 = 0$$
$$H_a : \mu_1 - \mu_2 \neq 0$$

- But the samples are not independent. The pairs of responses are linked because measurements are taken on the same car.

# The Paired-Difference Test

To test $H_0 : \mu_1 - \mu_2 = 0$ we test $H_0 : \mu_d = 0$ using the test statistic

$$t = \frac{\bar{d} - 0}{s_d / \sqrt{n}}$$

where $n$ = number of pairs, $\bar{d}$ and $s_d$ are the mean and standard deviation of the differences, $d_i$. Use the $p$ - value or a rejection region based on a t - distribution with $df = n - 1$.

# EXAMPLE

| Car | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Type A | 10.6 | 9.8 | 12.3 | 9.7 | 8.8 |
| Type B | 10.2 | 9.4 | 11.8 | 9.1 | 8.3 |
| Difference | .4 | .4 | .5 | .6 | .5 |

$$H_0 : \mu_1 - \mu_2 = 0$$
$$H_a : \mu_1 - \mu_2 \neq 0$$

Test statistic :

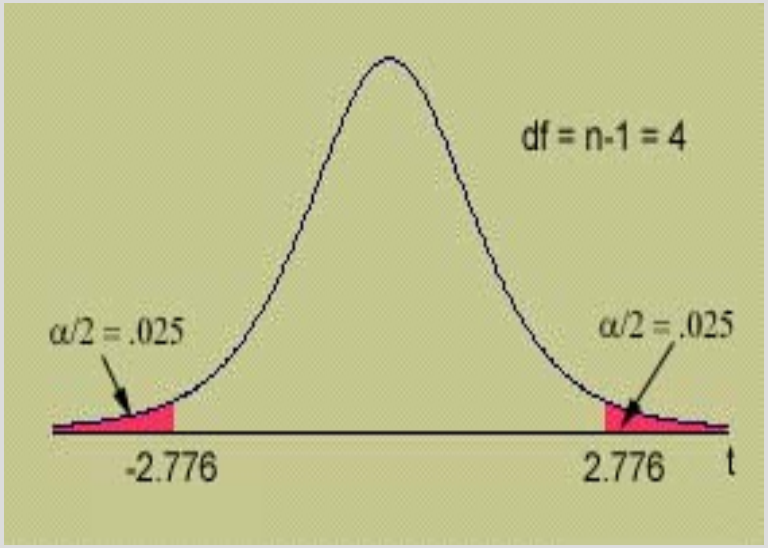
$$t = \frac{\bar{d} - 0}{s_d / \sqrt{n}} = \frac{.48 - 0}{.0837 / \sqrt{5}} = 12.8$$

Calculated $\bar{d} = \frac{\sum d_i}{n} = .48$

$$s_d = \sqrt{\frac{\sum d_i^2 - \frac{\left(\sum d_i\right)^2}{n}}{n-1}} = .0837$$

# Example

| Car | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Type A | 10.6 | 9.8 | 12.3 | 9.7 | 8.8 |
| Type B | 10.2 | 9.4 | 11.8 | 9.1 | 8.3 |
| Difference | .4 | .4 | .5 | .6 | .5 |




**<u>Rejection region</u>**: Reject $H_0$ if $t > 2.776$ or $t < -2.776$.

**<u>Conclusion:</u>** Since $t = 12.8$, $H_0$ is rejected. There is a difference in the average tire wear for the two types of tires.

# REJECTION REGION APPROACH

1. $H_0 : \mu_1 - \mu_2 = 0$ vs $H_a : \mu_1 - \mu_2 \neq 0$
2. Set $\alpha = 0.05$.
3. The test statistic and its sampling distribution is

$$t_{STAT} = \frac{\bar{D} - 0}{S_D/\sqrt{n}} \sim t_{(n-1)}\,.$$

4. Calculate the test statistic

$$t^* = \frac{\bar{d} - 0}{s_d/\sqrt{n}} = \frac{0.48 - 0}{0.0837/\sqrt{5}} = 12.8$$

5. Rejection region: $\{t : t > 2.776 \text{ or } t < -2.776\}$
6. Because $t^*$ falls in the rejection region, we reject $H_0$. We conclude that there seems to be a difference in the average tire wear for the two types of tires.

# $p$-VALUE APPROACH

5. The $p$-value is approximated by

$$p-\text{value} = 2P(t_4 > 12.8) < 2 \times 0.005 = 0.01$$

6. Because the $p$-value is less than $\alpha = 0.05$, we reject $H_0$. We conclude that there seems to be a difference in the average tire wear for the two types of tires.

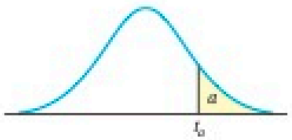
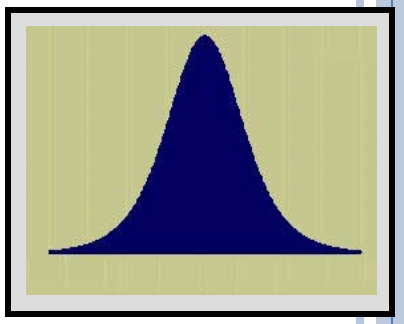



**Table 4 Critical Values of $t$**

| df | $t_{.100}$ | $t_{.050}$ | $t_{.025}$ | $t_{.010}$ | $t_{.005}$ | df |
|---|---|---|---|---|---|---|
| 1 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 | 1 |
| 2 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 | 2 |
| 3 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 | 3 |
| 4 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 | 4 |
| 5 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 | 5 |

# Some Notes



- Once you have designed the experiment by pairing, you MUST analyze it as a paired experiment. If the experiment is not designed as a paired experiment in advance, do not use this procedure.

# 10.5 INFERENCES CONCERNING A POPULATION VARIANCE

# Key ingrendients

1. Model: $X_i \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ for $i = 1, \ldots, n$.
2. Point estimator of $\sigma^2$: use sample variance $S^2$ to estimate $\sigma^2$.

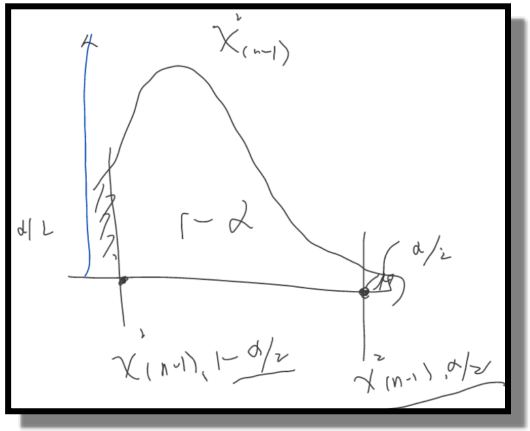
$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 .$$

3. Standardize the point estimator and find its distribution:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)} .$$

**An idea but not proof:** $\frac{(n-1)S^2}{\sigma^2} = \sum \left( \frac{X_i - \bar{X}}{\sigma} \right)^2 \approx \sum Z_i^2$

# A $100(1-\alpha)\,\%$ Confidencen interval for $\sigma^2$




- In a similar manner,

$$1-\alpha = P\left(\chi^2_{(n-1);1-\alpha/2} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{(n-1);\alpha/2}\right)$$

$$= P\left(\frac{1}{\chi^2_{(n-1);\alpha/2}} < \frac{\sigma^2}{(n-1)S^2} < \frac{1}{\chi^2_{(n-1);1-\alpha/2}}\right)$$

$$= P\left(\frac{(n-1)S^2}{\chi^2_{(n-1);\alpha/2}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{(n-1);1-\alpha/2}}\right).$$

- Thus, a $100(1-\alpha)\,\%$ CI for $\sigma^2$ is

$$\left[\frac{(n-1)s^2}{\chi^2_{(n-1),\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{(n-1),1-\alpha/2}})\right].$$

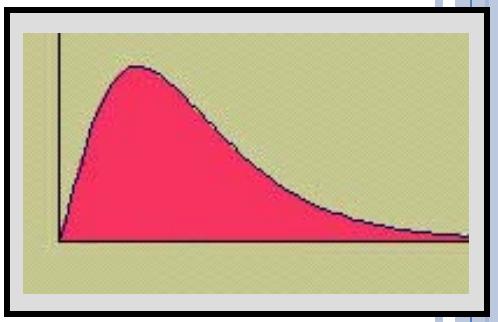
# Hypothesis test

1. $H_0 : \sigma^2 = \sigma_0^2$ v.s. $H_1 : \sigma^2 \neq (>, <)\sigma_0^2$.
2. Set $\alpha$.
3. Test statistic and its sampling distribution

$$\chi^2_{STAT} = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_{(n-1)}.$$

4. Calculate the realized statistic, $\chi^{2*} = \dfrac{(n-1)s^2}{\sigma_0^2}$.
5. Find the rejection region or $p$-value.
6. Conclude.

# Inference Concerning a Population Variance

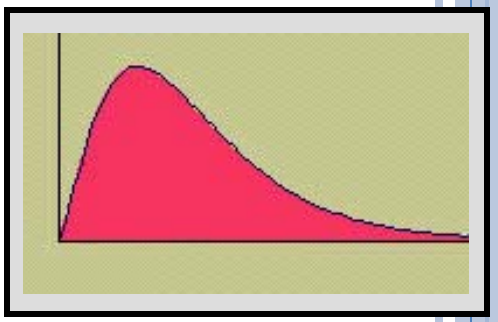


- Sometimes the primary parameter of interest is not the population mean $\mu$ but rather the population variance $\sigma^2$. We choose a random sample of size *n* from a normal distribution.
- The sample variance $s^2$ can be used in its standardized form:

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

- which has a Chi-Square distribution with *n* - 1 degrees of freedom.

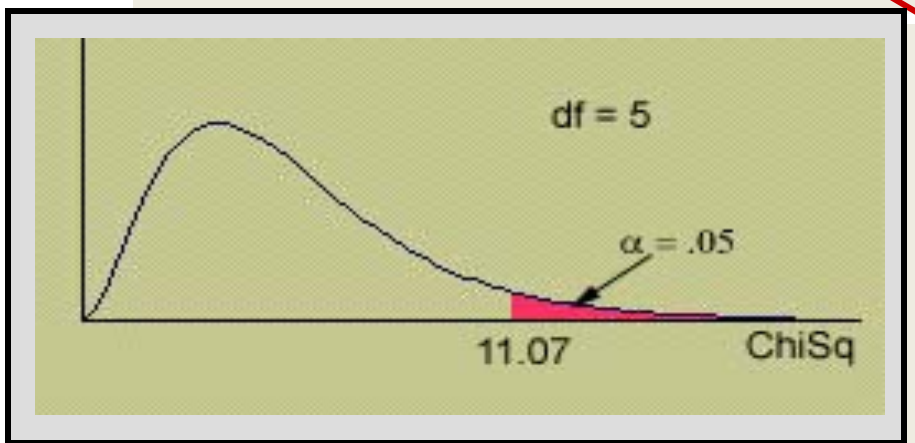
# Inference Concerning a Population Variance



- Table 5 gives both upper and lower critical values of the chi-square statistic for a given *df*.

TABLE 5 (continued)

| $\chi^2_{.100}$ | $\chi^2_{.050}$ | $\chi^2_{.025}$ | $\chi^2_{.010}$ | $\chi^2_{.005}$ | *df* |
|---|---|---|---|---|---|
| 2.70554 | 3.84146 | 5.02389 | 6.63490 | 7.87944 | 1 |
| 4.60517 | 5.99147 | 7.37776 | 9.21034 | 10.5966 | 2 |
| 6.25139 | 7.81473 | 9.34840 | 11.3449 | 12.8381 | 3 |
| 7.77944 | 9.48773 | 11.1433 | 13.2767 | 14.8602 | 4 |
| 9.23635 | 11.0705 | 12.8325 | 15.0863 | 16.7496 | 5 |
| 10.6446 | 12.5916 | 14.4494 | 16.8119 | 18.5476 | 6 |
| 12.0170 | 14.0671 | 16.0128 | 18.475[illegible] | [illegible] | [illegible] |




**For example, the value of chi-square that cuts off .05 in the upper tail of the distribution with *df* = 5 is $\chi^2$ =11.07.**

# EXAMPLE

- A cement manufacturer claims that his cement has a compressive strength with a standard deviation of 10 kg/cm$^2$ or less. A sample of $n$ = 10 measurements produced a mean and standard deviation of 312 and 13.96, respectively.

**A test of hypothesis:**

**$H_0$: $\sigma^2 = 10$ (claim is correct)**

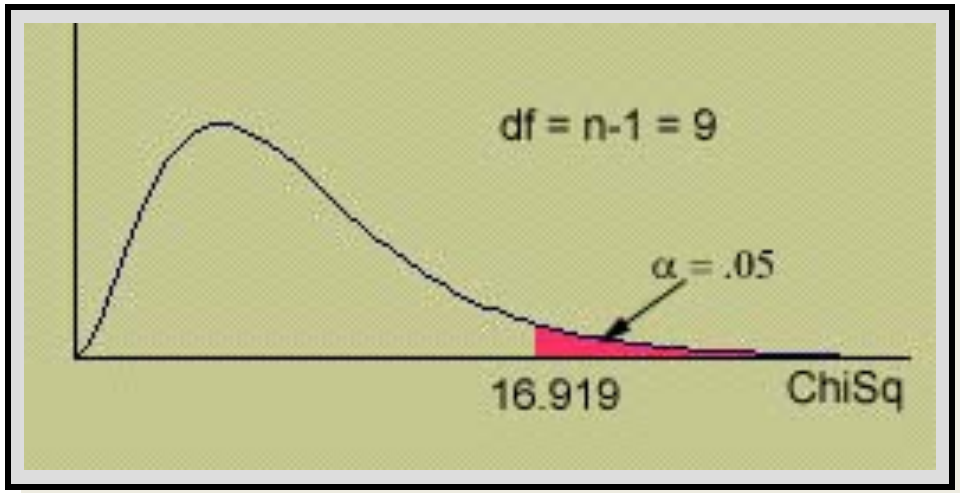
**$H_a$: $\sigma^2 > 10$ (claim is wrong)**

**uses the test statistic:**

$$\chi^2 = \frac{(n-1)s^2}{10^2} = \frac{9(13.96^2)}{100} = 17.5$$

# EXAMPLE

- Do these data produce sufficient evidence to reject the manufacturer's claim? Use $\alpha$ = .05.

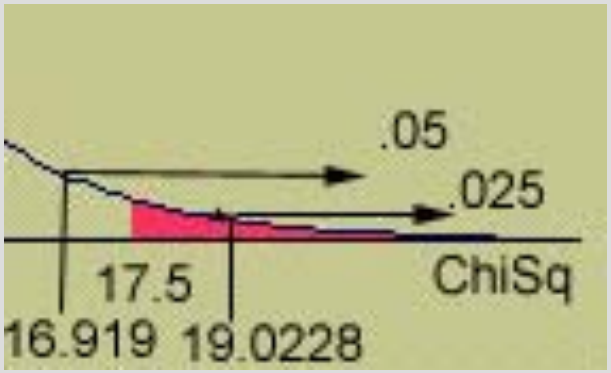



**<u>Rejection region</u>:**
Reject $H_0$ if $\chi^2 > 16.919$ ($\alpha = .05$).

**<u>Conclusion:</u>** Since $\chi^2 = 17.5$, $H_0$ is rejected. The standard deviation of the cement strengths is more than 10.

# APPROXIMATING THE *P*-VALUE

$$p\text{-value}: P(\chi^2 > 17.5) \text{ with } df = n - 1 = 9$$




| $\chi^2_{.100}$ | $\chi^2_{.050}$ | $\chi^2_{.025}$ | $\chi^2_{.010}$ | $\chi^2_{.005}$ | *df* |
|---|---|---|---|---|---|
| 2.70554 | 3.84146 | 5.02389 | 6.63490 | 7.87944 | 1 |
| 4.60517 | 5.99147 | 7.37776 | 9.21034 | 10.5966 | 2 |
| 6.25139 | 7.81473 | 9.34840 | 11.3449 | 12.8381 | 3 |
| 7.77944 | 9.48773 | 11.1433 | 13.2767 | 14.8602 | 4 |
| 9.23635 | 11.0705 | 12.8325 | 15.0863 | 16.7496 | 5 |
| 10.6446 | 12.5916 | 14.4494 | 16.8119 | 18.5476 | 6 |
| 12.0170 | 14.0671 | 16.0128 | 18.4753 | 20.2777 | 7 |
| 13.3616 | 15.5073 | 17.5346 | 20.0902 | 21.9550 | 8 |
| 14.6837 | 16.9190 | 19.0228 | 21.6660 | 23.5893 | 9 |

.025 < *p*-value < .05

Since the *p*-value is less than $\alpha$ = .05, $H_0$ is rejected. There is sufficient evidence to reject the manufacturer's claim.

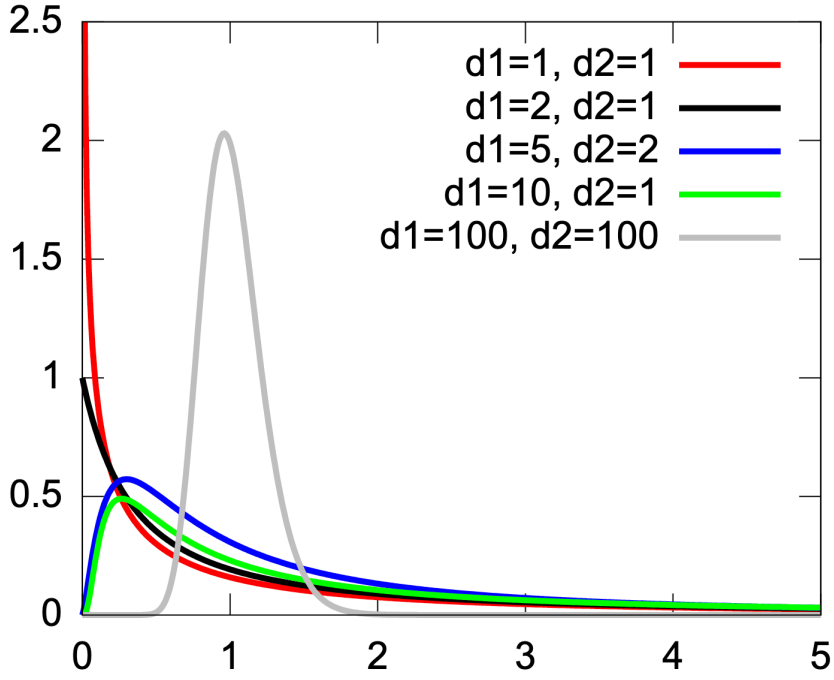
# STAT 10.6

## Comparing two population variances (Skipped)

# The $F$ distribution

- If $U \sim \chi^2_{\nu_1}$ and $V \sim \chi^2_{\nu_2}$, and they are independent. Then,

$$\frac{U/\nu_1}{V/\nu_2} \sim \frac{\chi^2_{\nu_1}/\nu_1}{\chi^2_{\nu_2}/\nu_2} \sim F_{\nu_1,\nu_2}$$



# SAMPLE VARIANCES

- $S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)^2, \quad S_2^2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (X_{2j} - \bar{X}_2)^2$
- $\bar{X}_1 = \frac{1}{n_1} \sum X_{1i}$
- $\bar{X}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} X_{2j}$

# Key ingredients

1. Model

$$X_{1i} \sim N(\mu_1, \sigma_1^2), \quad i = 1, \ldots, n_1$$

$$X_{2j} \sim N(\mu_2, \sigma_2^2), \quad j = 1, \ldots, n_2 \,.$$

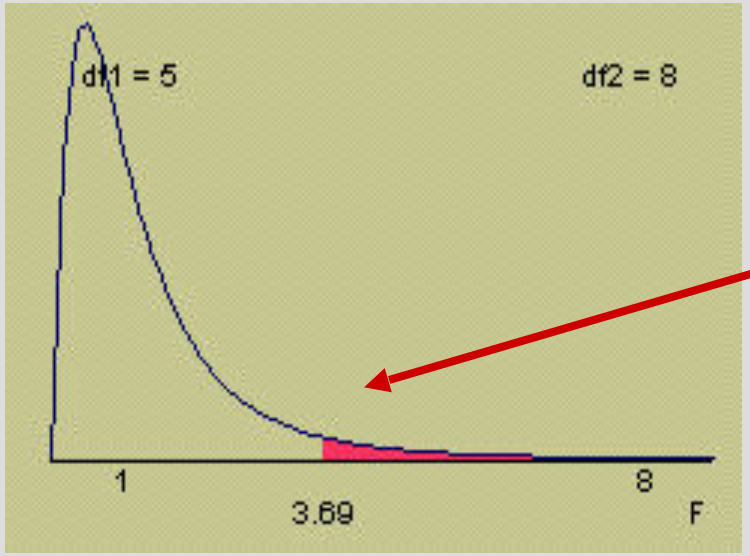
2. Point estimator for $\sigma_1^2/\sigma_2^2$ : $S_1^2/S_2^2$

3. Standardize the estimator and find its distribution:

$$\frac{\dfrac{(n_1-1)S_1^2/\sigma_1^2}{(n_1-1)}}{\dfrac{(n_2-1)S_2^2/\sigma_2^2}{(n_2-1)}} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{(n_1-1),(n_2-1)}$$

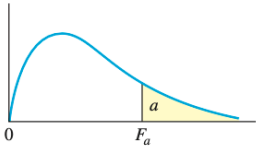
# Inference Concerning Two Population Variances

- Table 6 gives only upper critical values of the F statistic for a given pair of $df_1$ and $df_2$.




**For example, the value of F that cuts off .05 in the upper tail of the distribution with $df_1$**

## Table 6 Percentage Points of the *F* Distribution




| | | $df_1$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $df_2$ | $a$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | .100 | 39.86 | 49.50 | 53.59 | 55.83 | 57.24 | 58.20 | 58.91 | 59.44 | 59.86 |
| | .050 | 161.4 | 199.5 | 215.7 | 224.6 | 230.2 | 234.0 | 236.8 | 238.9 | 240.5 |
| | .025 | 647.8 | 799.5 | 864.2 | 899.6 | 921.8 | 937.1 | 948.2 | 956.7 | 963.3 |
| | .010 | 4052 | 4999.5 | 5403 | 5625 | 5764 | 5859 | 5928 | 5982 | 6022 |
| | .005 | 16211 | 20000 | 21615 | 22500 | 23056 | 23437 | 23715 | 23925 | 24091 |
| 2 | .100 | 8.53 | 9.00 | 9.16 | 9.24 | 9.29 | 9.33 | 9.35 | 9.37 | 9.38 |
| | .050 | 18.51 | 19.00 | 19.16 | 19.25 | 19.30 | 19.33 | 19.35 | 19.37 | 19.38 |
| | .025 | 38.51 | 39.00 | 39.17 | 39.25 | 39.30 | 39.33 | 39.36 | 39.37 | 39.39 |
| | .010 | 98.50 | 99.00 | 99.17 | 99.25 | 99.30 | 99.33 | 99.36 | 99.37 | 99.39 |
| | .005 | 198.5 | 199.0 | 199.2 | 199.2 | 199.3 | 199.3 | 199.4 | 199.4 | 199.4 |
| 3 | .100 | 5.54 | 5.46 | 5.39 | 5.34 | 5.31 | 5.28 | 5.27 | 5.25 | 5.24 |
| | .050 | 10.13 | 9.55 | 9.28 | 9.12 | 9.01 | 8.94 | 8.89 | 8.85 | 8.81 |
| | .025 | 17.44 | 16.04 | 15.44 | 15.10 | 14.88 | 14.73 | 14.62 | 14.54 | 14.47 |
| | .010 | 34.12 | 30.82 | 29.46 | 28.71 | 28.24 | 27.91 | 27.64 | 27.49 | 27.35 |
| | .005 | 55.55 | 49.80 | 47.47 | 46.19 | 45.39 | 44.84 | 44.43 | 44.13 | 43.88 |
| 4 | .100 | 4.54 | 4.32 | 4.19 | 4.11 | 4.05 | 4.01 | 3.98 | 3.95 | 3.94 |
| | .050 | 7.71 | 6.94 | 6.59 | 6.39 | 6.26 | 6.16 | 6.09 | 6.04 | 6.00 |
| | .025 | 12.22 | 10.65 | 9.98 | 9.60 | 9.36 | 9.20 | 9.07 | 8.98 | 8.90 |
| | .010 | 21.20 | 18.00 | 16.69 | 15.98 | 15.52 | 15.21 | 14.98 | 14.80 | 14.66 |
| | .005 | 31.33 | 26.28 | 24.26 | 23.15 | 22.46 | 21.97 | 21.62 | 21.35 | 21.14 |
| 5 | .100 | 4.06 | 3.78 | 3.62 | 3.52 | 3.45 | 3.40 | 3.37 | 3.34 | 3.32 |
| | .050 | 6.61 | 5.79 | 5.41 | 5.19 | 5.05 | 4.95 | 4.88 | 4.82 | 4.77 |
| | .025 | 10.01 | 8.43 | 7.76 | 7.39 | 7.15 | 6.98 | 6.85 | 6.76 | 6.68 |
| | .010 | 16.26 | 13.27 | 12.06 | 11.39 | 10.97 | 10.67 | 10.46 | 10.29 | 10.16 |
| | .005 | 22.78 | 18.31 | 16.53 | 15.56 | 14.94 | 14.51 | 14.20 | 13.96 | 13.77 |
| 6 | .100 | 3.78 | 3.46 | 3.29 | 3.18 | 3.11 | 3.05 | 3.01 | 2.98 | 2.96 |
| | .050 | 5.99 | 5.14 | 4.76 | 4.53 | 4.39 | 4.28 | 4.21 | 4.15 | 4.10 |
| | .025 | 8.81 | 7.26 | 6.60 | 6.23 | 5.99 | 5.82 | 5.70 | 5.60 | 5.52 |
| | .010 | 13.75 | 10.92 | 9.78 | 9.15 | 8.75 | 8.47 | 8.26 | 8.10 | 7.98 |
| | .005 | 18.63 | 14.54 | 12.92 | 12.03 | 11.46 | 11.07 | 10.79 | 10.57 | 10.39 |
| 7 | .100 | 3.59 | 3.26 | 3.07 | 2.96 | 2.88 | 2.83 | 2.78 | 2.75 | 2.72 |
| | .050 | 5.59 | 4.74 | 4.35 | 4.12 | 3.97 | 3.87 | 3.79 | 3.73 | 3.68 |
| | .025 | 8.07 | 6.54 | 5.89 | 5.52 | 5.29 | 5.12 | 4.99 | 4.90 | 4.82 |
| | .010 | 12.25 | 9.55 | 8.45 | 7.85 | 7.46 | 7.19 | 6.99 | 6.84 | 6.72 |
| | .005 | 16.24 | 12.40 | 10.88 | 10.05 | 9.52 | 9.16 | 8.89 | 8.68 | 8.51 |
| 8 | .100 | 3.46 | 3.11 | 2.92 | 2.81 | 2.73 | 2.67 | 2.62 | 2.59 | 2.56 |
| | .050 | 5.32 | 4.46 | 4.07 | 3.84 | 3.69 | 3.58 | 3.50 | 3.44 | 3.39 |
| | .025 | 7.57 | 6.06 | 5.42 | 5.05 | 4.82 | 4.65 | 4.53 | 4.43 | 4.36 |
| | .010 | 11.26 | 8.65 | 7.59 | 7.01 | 6.63 | 6.37 | 6.18 | 6.03 | 5.91 |
| | .005 | 14.69 | 11.04 | 9.60 | 8.81 | 8.30 | 7.95 | 7.69 | 7.50 | 7.34 |
| 9 | .100 | 3.36 | 3.01 | 2.81 | 2.69 | 2.61 | 2.55 | 2.51 | 2.47 | 2.44 |
| | .050 | 5.12 | 4.26 | 3.86 | 3.63 | 3.48 | 3.37 | 3.29 | 3.23 | 3.18 |
| | .025 | 7.21 | 5.71 | 5.08 | 4.72 | 4.48 | 4.32 | 4.20 | 4.10 | 4.03 |
| | .010 | 10.56 | 8.02 | 6.99 | 6.42 | 6.06 | 5.80 | 5.61 | 5.47 | 5.35 |
| | .005 | 13.61 | 10.11 | 8.72 | 7.96 | 7.47 | 7.13 | 6.88 | 6.69 | 6.54 |

# CI FOR $\sigma_1^2/\sigma_2^2$

- The $100(1-\alpha)\%$ CI for $\sigma_1^2/\sigma_2^2$

$$\left[\frac{s_1^2}{s_2^2}\frac{1}{F_{\nu_1,\nu_2;\alpha/2}}, \frac{s_1^2}{s_2^2}F_{\nu_2,\nu_1;\alpha/2}\right]$$

- Why?

$$\begin{aligned}
1-\alpha &= P\left(F_{\nu_1,\nu_2;1-\alpha/2} < \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} < F_{\nu_1,\nu_2;\alpha/2}\right) \\
&= P\left(F_{\nu_1,\nu_2;1-\alpha/2} < \frac{S_1^2}{S_2^2}\frac{\sigma_2^2}{\sigma_1^2} < F_{\nu_1,\nu_2;\alpha/2}\right) \\
&= P\left(\frac{1}{F_{\nu_1,\nu_2;\alpha/2}} < \frac{1}{S_1^2/S_2^2}\frac{\sigma_1^2}{\sigma_2^2} < \frac{1}{F_{\nu_1,\nu_2;1-\alpha/2}}\right) \\
&= P\left(\frac{S_1^2}{S_2^2}\frac{1}{F_{\nu_1,\nu_2;\alpha/2}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2}\frac{1}{F_{\nu_1,\nu_2;1-\alpha/2}}\right) \\
&= P\left(\frac{S_1^2}{S_2^2}\frac{1}{F_{\nu_1,\nu_2;\alpha/2}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2}F_{\nu_2,\nu_1;\alpha/2}\right).
\end{aligned}$$

# Why $F_{\nu_1,\nu_2;1-\alpha} = \dfrac{1}{F_{\nu_2,\nu_1;\alpha}}$ ?

Note $F_{\nu_1,\nu_2} \sim \dfrac{\chi^2_{\nu_1}/\nu_1}{\chi^2_{\nu_2}/\nu_2} = \dfrac{1}{\dfrac{\chi^2_{\nu_2}/\nu_2}{\chi^2_{\nu_1}/\nu_1}} \sim \dfrac{1}{F_{\nu_2,\nu_1}}$

1. First, $\alpha = \boxed{P(F_{\nu_2,\nu_1} > \boxed{F_{\nu_2,\nu_1;\alpha}})}$
2. Second,

$$\alpha = (F_{\nu_1,\nu_2} < F_{\nu_1,\nu_2,1-\alpha}) = P(\frac{1}{F_{\nu_2,\nu_1}} < F_{\nu_1,\nu_2,1-\alpha}) = \boxed{P(F_{\nu_2,\nu_1} > \boxed{\frac{1}{F_{\nu_1,\nu_2;1-\alpha}}})}$$

Thus, they must be the same.

# Inference Concerning Two Population Variances

- We can make inferences about the ratio of two population variances in the form a ratio. We choose two independent random samples of size $n_1$ and $n_2$ from normal distributions.
- If the two population variances are equal, the statistic

$$F = \frac{s_1^2}{s_2^2}$$

- has an $F$ distribution with $df_1 = n_1 - 1$ and $df_2 = n_2 - 1$ degrees of freedom.

# EXAMPLE

- An experimenter has performed a experiment using two groups of rats. He wants to test $H_0$: $\mu_1 = \mu_2$, but first he wants to make sure that the population variances are equal.

| | Standard (2) | Experimental (1) |
|---|---|---|
| Sample size | 10 | 11 |
| Sample mean | 13.64 | 12.42 |
| Sample Std Dev | 2.3 | 5.8 |

Find the 95% CI for $\sigma_1^2/\sigma_2^2$!

■ Table 6 (continued)

| $df_1$ | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 10 | 12 | 15 | 20 | 24 | 30 | 40 | 60 | 120 | ∞ | $\alpha$ | $df_2$ |
| 60.19 | 60.71 | 61.22 | 61.74 | 62.00 | 62.26 | 62.53 | 62.79 | 63.06 | 63.33 | .100 | 1 |
| 241.9 | 243.9 | 245.9 | 248.0 | 249.1 | 250.1 | 251.2 | 252.2 | 253.3 | 254.3 | .050 | |
| 968.6 | 976.7 | 984.9 | 993.1 | 997.2 | 1001 | 1006 | 1010 | 1014 | 1018 | .025 | |
| 6056 | 6106 | 6157 | 6209 | 6235 | 6261 | 6287 | 6313 | 6339 | 6366 | .010 | |
| 24224 | 24426 | 24630 | 24836 | 24940 | 25044 | 25148 | 25253 | 25359 | 25465 | .005 | |
| 9.39 | 9.41 | 9.42 | 9.44 | 9.45 | 9.46 | 9.47 | 9.47 | 9.48 | 9.49 | .100 | 2 |
| 19.40 | 19.41 | 19.43 | 19.45 | 19.45 | 19.46 | 19.47 | 19.48 | 19.49 | 19.50 | .050 | |
| 39.40 | 39.41 | 39.43 | 39.45 | 39.46 | 39.46 | 39.47 | 39.48 | 39.49 | 39.50 | .025 | |
| 99.40 | 99.42 | 99.43 | 99.45 | 99.46 | 99.47 | 99.47 | 99.48 | 99.49 | 99.50 | .010 | |
| 199.4 | 199.4 | 199.4 | 199.4 | 199.5 | 199.5 | 199.5 | 199.5 | 199.5 | 199.5 | .005 | |
| 5.23 | 5.22 | 5.20 | 5.18 | 5.18 | 5.17 | 5.16 | 5.15 | 5.14 | 5.13 | .100 | 3 |
| 8.79 | 8.74 | 8.70 | 8.66 | 8.64 | 8.62 | 8.59 | 8.57 | 8.55 | 8.53 | .050 | |
| 14.42 | 14.34 | 14.25 | 14.17 | 14.12 | 14.08 | 14.04 | 13.99 | 13.95 | 13.90 | .025 | |
| 27.23 | 27.05 | 26.87 | 26.69 | 26.60 | 26.50 | 26.41 | 26.32 | 26.22 | 26.13 | .010 | |
| 43.69 | 43.39 | 43.08 | 42.78 | 42.62 | 42.47 | 42.31 | 42.15 | 41.99 | 41.83 | .005 | |
| 3.92 | 3.90 | 3.87 | 3.84 | 3.83 | 3.82 | 3.80 | 3.79 | 3.78 | 3.76 | .100 | 4 |
| 5.96 | 5.91 | 5.86 | 5.80 | 5.77 | 5.75 | 5.72 | 5.69 | 5.66 | 5.63 | .050 | |
| 8.84 | 8.75 | 8.66 | 8.56 | 8.51 | 8.46 | 8.41 | 8.36 | 8.31 | 8.26 | .025 | |
| 14.55 | 14.37 | 14.20 | 14.02 | 13.93 | 13.84 | 13.75 | 13.65 | 13.56 | 13.46 | .010 | |
| 20.97 | 20.70 | 20.44 | 20.17 | 20.03 | 19.89 | 19.75 | 19.61 | 19.47 | 19.32 | .005 | |
| 3.30 | 3.27 | 3.24 | 3.21 | 3.19 | 3.17 | 3.16 | 3.14 | 3.12 | 3.10 | .100 | 5 |
| 4.74 | 4.68 | 4.62 | 4.56 | 4.53 | 4.50 | 4.46 | 4.43 | 4.40 | 4.36 | .050 | |
| 6.62 | 6.52 | 6.43 | 6.33 | 6.28 | 6.23 | 6.18 | 6.12 | 6.07 | 6.02 | .025 | |
| 10.05 | 9.89 | 9.72 | 9.55 | 9.47 | 9.38 | 9.29 | 9.20 | 9.11 | 9.02 | .010 | |
| 13.62 | 13.38 | 13.15 | 12.90 | 12.78 | 12.66 | 12.53 | 12.40 | 12.27 | 12.14 | .005 | |
| 2.94 | 2.90 | 2.87 | 2.84 | 2.82 | 2.80 | 2.78 | 2.76 | 2.74 | 2.72 | .100 | 6 |
| 4.06 | 4.00 | 3.94 | 3.87 | 3.84 | 3.81 | 3.77 | 3.74 | 3.70 | 3.67 | .050 | |
| 5.46 | 5.37 | 5.27 | 5.17 | 5.12 | 5.07 | 5.01 | 4.96 | 4.90 | 4.85 | .025 | |
| 7.87 | 7.72 | 7.56 | 7.40 | 7.31 | 7.23 | 7.14 | 7.06 | 6.97 | 6.88 | .010 | |
| 10.25 | 10.03 | 9.81 | 9.59 | 9.47 | 9.36 | 9.24 | 9.12 | 9.00 | 8.88 | .005 | |
| 2.70 | 2.67 | 2.63 | 2.59 | 2.58 | 2.56 | 2.54 | 2.51 | 2.49 | 2.47 | .100 | 7 |
| 3.64 | 3.57 | 3.51 | 3.44 | 3.41 | 3.38 | 3.34 | 3.30 | 3.27 | 3.23 | .050 | |
| 4.76 | 4.67 | 4.57 | 4.47 | 4.42 | 4.36 | 4.31 | 4.25 | 4.20 | 4.14 | .025 | |
| 6.62 | 6.47 | 6.31 | 6.16 | 6.07 | 5.99 | 5.91 | 5.82 | 5.74 | 5.65 | .010 | |
| 8.38 | 8.18 | 7.97 | 7.75 | 7.65 | 7.53 | 7.42 | 7.31 | 7.19 | 7.08 | .005 | |
| 2.54 | 2.50 | 2.46 | 2.42 | 2.40 | 2.38 | 2.36 | 2.34 | 2.32 | 2.29 | .100 | 8 |
| 3.35 | 3.28 | 3.22 | 3.15 | 3.12 | 3.08 | 3.04 | 3.01 | 2.97 | 2.93 | .050 | |
| 4.30 | 4.20 | 4.10 | 4.00 | 3.95 | 3.89 | 3.84 | 3.78 | 3.73 | 3.67 | .025 | |
| 5.81 | 5.67 | 5.52 | 5.36 | 5.28 | 5.20 | 5.12 | 5.03 | 4.95 | 4.86 | .010 | |
| 7.21 | 7.01 | 6.81 | 6.61 | 6.50 | 6.40 | 6.29 | 6.18 | 6.06 | 5.95 | .005 | |
| 2.42 | 2.38 | 2.34 | 2.30 | 2.28 | 2.25 | 2.23 | 2.21 | 2.18 | 2.16 | .100 | 9 |
| 3.14 | 3.07 | 3.01 | 2.94 | 2.90 | 2.86 | 2.83 | 2.79 | 2.75 | 2.71 | .050 | |
| 3.96 | 3.87 | 3.77 | 3.67 | 3.61 | 3.56 | 3.51 | 3.45 | 3.39 | 3.33 | .025 | |
| 5.26 | 5.11 | 4.96 | 4.81 | 4.73 | 4.65 | 4.57 | 4.48 | 4.40 | 4.31 | .010 | |
| 6.42 | 6.23 | 6.03 | 5.83 | 5.73 | 5.62 | 5.52 | 5.41 | 5.30 | 5.19 | .005 | |

■ Table 6 (continued)

| | | $df_1$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $df_2$ | $\alpha$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 10 | .100 | 3.29 | 2.92 | 2.73 | 2.61 | 2.52 | 2.46 | 2.41 | 2.38 | 2.35 |
| | .050 | 4.96 | 4.10 | 3.71 | 3.48 | 3.33 | 3.22 | 3.14 | 3.07 | 3.02 |
| | .025 | 6.94 | 5.46 | 4.83 | 4.47 | 4.24 | 4.07 | 3.95 | 3.85 | 3.78 |
| | .010 | 10.04 | 7.56 | 6.55 | 5.99 | 5.64 | 5.39 | 5.20 | 5.06 | 4.94 |
| | .005 | 12.83 | 9.43 | 8.08 | 7.34 | 6.87 | 6.54 | 6.30 | 6.12 | 5.97 |

# 95% CI FOR $\sigma_1^2/\sigma_2^2$

- Check $F$ table:

$$F_{10,9;0.025} = 3.96, \text{ and } F_{9,10;0.025} = 3.78$$

- The 95% CI for $\sigma_1^2/\sigma_2^2$:

$$\left[\frac{s_1^2}{s_2^2}\frac{1}{F_{\nu_1,\nu_2;\alpha/2}}, \frac{s_1^2}{s_2^2}F_{\nu_2,\nu_1;\alpha/2}\right] = \left[\frac{5.8^2}{2.3^2}\frac{1}{3.96} \quad \frac{5.8^2}{2.3^2}3.78\right] = [1.61, \quad 24.04]$$

- Because the above CI does not overlap 1, the variances should be different.

# EXAMPLE

- An experimenter has performed a experiment using two groups of rats. He wants to test $H_0$: $\mu_1 = \mu_2$, but first he wants to make sure that the population variances are equal.

| | **Standard (2)** | **Experimental (1)** |
|---|---|---|
| Sample size | 10 | 11 |
| Sample mean | 13.64 | 12.42 |
| Sample Std Dev | 2.3 | 5.8 |

Preliminary test:

$$H_0 : \sigma_1^2 = \sigma_2^2 \text{ versus } H_a : \sigma_1^2 \neq \sigma_2^2$$

# Hypothesis test

- $H_0 : \dfrac{\sigma_1^2}{\sigma_2^2} = \dfrac{\sigma_{1,0}^2}{\sigma_{2,0}^2}$ vs $H_1 : \dfrac{\sigma_1^2}{\sigma_2^2} \neq \dfrac{\sigma_{1,0}^2}{\sigma_{2,0}^2}$
- The test statistic and its sampling distribution is

$$F_{STAT} = \frac{S_1^2/\sigma_{1,0}^2}{S_2^2/\sigma_{2,0}^2} \sim F_{(n_1-1),(n_2-1)}$$

# Hypothesis test

1. $H_0 : \dfrac{\sigma_1^2}{\sigma_2^2} = 1$ vs $H_a : \dfrac{\sigma_1^2}{\sigma_2^2} \neq 1$
2. Set $\alpha$
3. Test statistic: $F_{STAT} = \dfrac{S_1^2}{S_2^2} \sim F_{(n_1-1),(n_2-1)}$
4. Calculate the realized statistic $F^* = \dfrac{s_1^2}{s_2^2}$
5. Find the rejection region or p-value
6. Conclude

A trick for two-sided test: we designate the sample with the larger standard deviation as sample 1, to force the test statistic into the upper tail of the *F* distribution!

# EXAMPLE

| | Standard (2) | Experimental (1) |
|---|---|---|
| Sample size | 10 | 11 |
| Sample Std Dev | 2.3 | 5.8 |

$$H_0 : \sigma_1^2 = \sigma_2^2$$
$$H_a : \sigma_1^2 \neq \sigma_2^2$$

Test statistic :

$$F = \frac{s_1^2}{s_2^2} = \frac{5.8^2}{2.3^2} = 6.36$$

We designate the sample with the larger standard deviation as sample 1, to force the test statistic into the upper tail of the *F* distribution.

# EXAMPLE

$$H_0 : \sigma_1^2 = \sigma_2^2$$
$$H_a : \sigma_1^2 \neq \sigma_2^2$$

Test statistic :

$$F = \frac{s_1^2}{s_2^2} = \frac{5.8^2}{2.3^2} = 6.36$$

The rejection region is two-tailed, with $\alpha$ = .05, but we only need to find the upper critical value, which has $\alpha/2$ = .025 to its right.

From Table 6, with $df_1$=10 and $df_2$ = 9, we reject $H_0$ if $F$ > 3.96.

**CONCLUSION:** Reject $H_0$. There is sufficient evidence to indicate that the variances are **unequal.** Do not rely on the assumption of equal variances for your *t test!*

# Rejection region approach

1. $H_0 : \dfrac{\sigma_1^2}{\sigma_2^2} = 1$ vs $H_a : \dfrac{\sigma_1^2}{\sigma_2^2} \neq 1$
2. $\alpha = 0.05$
3. Test statistic: $F_{STAT} = \dfrac{S_1^2}{S_2^2} \sim F_{10,9}$
4. Realized test statistic: $F^* = \dfrac{s_1^2}{s_2^2} = \dfrac{5.8^2}{2.3^2} = 6.36.$
5. The rejection region: $\{F : F > 3.96\}$ with $F_{10,9;0.025} = 3.96$

   Without the trick, the rejection region is $\{F : F < 0.3 \text{ or } F > 3.96\}$, with $F_{10,9;0.975} = 1/F_{10,9;0.025} = 1/3.78 = 0.3$ and $F_{10,9;0.025} = 3.96$.
6. Because $F^*$ falls into the rejection region, we reject $H_0$. We conclude that there is a statistically significant evidence that these two variances are unequal.

■ **Table 6** (continued)

| | | $df_1$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $df_2$ | $\alpha$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 10 | .100 | 3.29 | 2.92 | 2.73 | 2.61 | 2.52 | 2.46 | 2.41 | 2.38 | 2.35 |
| | .050 | 4.96 | 4.10 | 3.71 | 3.48 | 3.33 | 3.22 | 3.14 | 3.07 | 3.02 |
| | .025 | 6.94 | 5.46 | 4.83 | 4.47 | 4.24 | 4.07 | 3.95 | 3.85 | 3.78 |
| | .010 | 10.04 | 7.56 | 6.55 | 5.99 | 5.64 | 5.39 | 5.20 | 5.06 | 4.94 |
| | .005 | 12.83 | 9.43 | 8.08 | 7.34 | 6.87 | 6.54 | 6.30 | 6.12 | 5.97 |

■ **Table 6** (continued)

| $df_1$ | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 10 | 12 | 15 | 20 | 24 | 30 | 40 | 60 | 120 | ∞ | $\alpha$ | $df_2$ |
| 60.19 | 60.71 | 61.22 | 61.74 | 62.00 | 62.26 | 62.53 | 62.79 | 63.06 | 63.33 | .100 | 1 |
| 241.9 | 243.9 | 245.9 | 248.0 | 249.1 | 250.1 | 251.2 | 252.2 | 253.3 | 254.3 | .050 | |
| 968.6 | 976.7 | 984.9 | 993.1 | 997.2 | 1001 | 1006 | 1010 | 1014 | 1018 | .025 | |
| 6056 | 6106 | 6157 | 6209 | 6235 | 6261 | 6287 | 6313 | 6339 | 6366 | .010 | |
| 24224 | 24426 | 24630 | 24836 | 24940 | 25044 | 25148 | 25253 | 25359 | 25465 | .005 | |
| 9.39 | 9.41 | 9.42 | 9.44 | 9.45 | 9.46 | 9.47 | 9.47 | 9.48 | 9.49 | .100 | 2 |
| 19.40 | 19.41 | 19.43 | 19.45 | 19.45 | 19.46 | 19.47 | 19.48 | 19.49 | 19.50 | .050 | |
| 39.40 | 39.41 | 39.43 | 39.45 | 39.46 | 39.46 | 39.47 | 39.48 | 39.49 | 39.50 | .025 | |
| 99.40 | 99.42 | 99.43 | 99.45 | 99.46 | 99.47 | 99.47 | 99.48 | 99.49 | 99.50 | .010 | |
| 199.4 | 199.4 | 199.4 | 199.4 | 199.5 | 199.5 | 199.5 | 199.5 | 199.5 | 199.5 | .005 | |
| 5.23 | 5.22 | 5.20 | 5.18 | 5.18 | 5.17 | 5.16 | 5.15 | 5.14 | 5.13 | .100 | 3 |
| 8.79 | 8.74 | 8.70 | 8.66 | 8.64 | 8.62 | 8.59 | 8.57 | 8.55 | 8.53 | .050 | |
| 14.42 | 14.34 | 14.25 | 14.17 | 14.12 | 14.08 | 14.04 | 13.99 | 13.95 | 13.90 | .025 | |
| 27.23 | 27.05 | 26.87 | 26.69 | 26.60 | 26.50 | 26.41 | 26.32 | 26.22 | 26.13 | .010 | |
| 43.69 | 43.39 | 43.08 | 42.78 | 42.62 | 42.47 | 42.31 | 42.15 | 41.99 | 41.83 | .005 | |
| 3.92 | 3.90 | 3.87 | 3.84 | 3.83 | 3.82 | 3.80 | 3.79 | 3.78 | 3.76 | .100 | 4 |
| 5.96 | 5.91 | 5.86 | 5.80 | 5.77 | 5.75 | 5.72 | 5.69 | 5.66 | 5.63 | .050 | |
| 8.84 | 8.75 | 8.66 | 8.56 | 8.51 | 8.46 | 8.41 | 8.36 | 8.31 | 8.26 | .025 | |
| 14.55 | 14.37 | 14.20 | 14.02 | 13.93 | 13.84 | 13.75 | 13.65 | 13.56 | 13.46 | .010 | |
| 20.97 | 20.70 | 20.44 | 20.17 | 20.03 | 19.89 | 19.75 | 19.61 | 19.47 | 19.32 | .005 | |
| 3.30 | 3.27 | 3.24 | 3.21 | 3.19 | 3.17 | 3.16 | 3.14 | 3.12 | 3.10 | .100 | 5 |
| 4.74 | 4.68 | 4.62 | 4.56 | 4.53 | 4.50 | 4.46 | 4.43 | 4.40 | 4.36 | .050 | |
| 6.62 | 6.52 | 6.43 | 6.33 | 6.28 | 6.23 | 6.18 | 6.12 | 6.07 | 6.02 | .025 | |
| 10.05 | 9.89 | 9.72 | 9.55 | 9.47 | 9.38 | 9.29 | 9.20 | 9.11 | 9.02 | .010 | |
| 13.62 | 13.38 | 13.15 | 12.90 | 12.78 | 12.66 | 12.53 | 12.40 | 12.27 | 12.14 | .005 | |
| 2.94 | 2.90 | 2.87 | 2.84 | 2.82 | 2.80 | 2.78 | 2.76 | 2.74 | 2.72 | .100 | 6 |
| 4.06 | 4.00 | 3.94 | 3.87 | 3.84 | 3.81 | 3.77 | 3.74 | 3.70 | 3.67 | .050 | |
| 5.46 | 5.37 | 5.27 | 5.17 | 5.12 | 5.07 | 5.01 | 4.96 | 4.90 | 4.85 | .025 | |
| 7.87 | 7.72 | 7.56 | 7.40 | 7.31 | 7.23 | 7.14 | 7.06 | 6.97 | 6.88 | .010 | |
| 10.25 | 10.03 | 9.81 | 9.59 | 9.47 | 9.36 | 9.24 | 9.12 | 9.00 | 8.88 | .005 | |
| 2.70 | 2.67 | 2.63 | 2.59 | 2.58 | 2.56 | 2.54 | 2.51 | 2.49 | 2.47 | .100 | 7 |
| 3.64 | 3.57 | 3.51 | 3.44 | 3.41 | 3.38 | 3.34 | 3.30 | 3.27 | 3.23 | .050 | |
| 4.76 | 4.67 | 4.57 | 4.47 | 4.42 | 4.36 | 4.31 | 4.25 | 4.20 | 4.14 | .025 | |
| 6.62 | 6.47 | 6.31 | 6.16 | 6.07 | 5.99 | 5.91 | 5.82 | 5.74 | 5.65 | .010 | |
| 8.38 | 8.18 | 7.97 | 7.75 | 7.65 | 7.53 | 7.42 | 7.31 | 7.19 | 7.08 | .005 | |
| 2.54 | 2.50 | 2.46 | 2.42 | 2.40 | 2.38 | 2.36 | 2.34 | 2.32 | 2.29 | .100 | 8 |
| 3.35 | 3.28 | 3.22 | 3.15 | 3.12 | 3.08 | 3.04 | 3.01 | 2.97 | 2.93 | .050 | |
| 4.30 | 4.20 | 4.10 | 4.00 | 3.95 | 3.89 | 3.84 | 3.78 | 3.73 | 3.67 | .025 | |
| 5.81 | 5.67 | 5.52 | 5.36 | 5.28 | 5.20 | 5.12 | 5.03 | 4.95 | 4.86 | .010 | |
| 7.21 | 7.01 | 6.81 | 6.61 | 6.50 | 6.40 | 6.29 | 6.18 | 6.06 | 5.95 | .005 | |
| 2.42 | 2.38 | 2.34 | 2.30 | 2.28 | 2.25 | 2.23 | 2.21 | 2.18 | 2.16 | .100 | 9 |
| 3.14 | 3.07 | 3.01 | 2.94 | 2.90 | 2.86 | 2.83 | 2.79 | 2.75 | 2.71 | .050 | |
| 3.96 | 3.87 | 3.77 | 3.67 | 3.61 | 3.56 | 3.51 | 3.45 | 3.39 | 3.33 | .025 | |
| 5.26 | 5.11 | 4.96 | 4.81 | 4.73 | 4.65 | 4.57 | 4.48 | 4.40 | 4.31 | .010 | |
| 6.42 | 6.23 | 6.03 | 5.83 | 5.73 | 5.62 | 5.52 | 5.41 | 5.30 | 5.19 | .005 | |

# P-VALUE APPROACH

5. Approximate the p-value

$$0.01 = 2 \times 0.005 < p\text{−value} < 2 \times 0.01 = 0.02$$

6. Because $p\text{−value} < \alpha$, we reject $H_0$. We conclude that there is a statistically significant evidence that these two variances are unequal

# More…

1. $H_0 : \dfrac{\sigma_1^2}{\sigma_2^2} = 5$ vs $H_a : \dfrac{\sigma_1^2}{\sigma_2^2} \neq 5$

2. $\alpha = 0.05$

3. Test statistic: $F_{STAT} = \dfrac{S_1^2}{S_2^2}\dfrac{1}{5} \sim F_{10,9}$

# 10.7

## Kay concepts

# Key Concepts

| Parameter | Test Statistic | Degrees of Freedom |
|---|---|---|
| $\mu$ | $t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ | $n - 1$ |
| $\mu_1 - \mu_2$ (equal variances) | $t = \dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$ | $n_1 + n_2 - 2$ |
| $\mu_1 - \mu_2$ (unequal variances) | $t \approx \dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$ | Satterthwaite's approximation |
| $\mu_1 - \mu_2$ (paired samples) | $t = \dfrac{\bar{d} - \mu_d}{s_d/\sqrt{n}}$ | $n - 1$ |
| $\sigma^2$ | $\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$ | $n - 1$ |
| $\sigma_1^2/\sigma_2^2$ | $F = s_1^2/s_2^2$ | $n_1 - 1$ and $n_2 - 1$ |

# Key Concepts

## I. Experimental Designs for Small Samples

1. Single random sample: The sampled population must be normal.
2. Two independent random samples: Both sampled populations must be normal.
   a. Populations have a common variance $\sigma^2$.
   b. Populations have different variances
3. Paired-difference or matched-pairs design: The samples are not independent.

# Key Concepts

## II. Statistical Tests of Significance

1. Based on the $t$, $F$, and $\chi^2$ distributions
2. Use the same procedure as in Chapter 9
3. Rejection region—critical values and significance levels: based on the $t$, $F$, and $\chi^2$ distributions with the appropriate degrees of freedom
4. Tests of population parameters: a single mean, the difference between two means, a single variance, and the ratio of two variances

## III. Small Sample Test Statistics

To test one of the population parameters when the sample sizes are small, use the following test statistics:

# Key Concepts

| Parameter | Test Statistic | Degrees of Freedom |
|---|---|---|
| $\mu$ | $t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ | $n - 1$ |
| $\mu_1 - \mu_2$ (equal variances) | $t = \dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$ | $n_1 + n_2 - 2$ |
| $\mu_1 - \mu_2$ (unequal variances) | $t \approx \dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$ | Satterthwaite's approximation |
| $\mu_1 - \mu_2$ (paired samples) | $t = \dfrac{\bar{d} - \mu_d}{s_d/\sqrt{n}}$ | $n - 1$ |
| $\sigma^2$ | $\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$ | $n - 1$ |
| $\sigma_1^2/\sigma_2^2$ | $F = s_1^2/s_2^2$ | $n_1 - 1$ and $n_2 - 1$ |